

MACHINE LEARNING WITH PYTHON ASSESSMENT

Wireless sound control

ABSTRACT :

Machine Learning is making the *computer learn from studying data and statistics*. Machine Learning is a step into the direction of Artificial Intelligence (AI). Machine Learning is a program that analyses data and learns to predict the outcome.

Python offers concise and readable code. While complex algorithms and versatile workflows stand behind machine learning and AI , Python's simplicity *allows developers to write reliable systems*. Developers get to put all their effort into solving an machine learning problem instead of focusing on the technical nuances of the language.

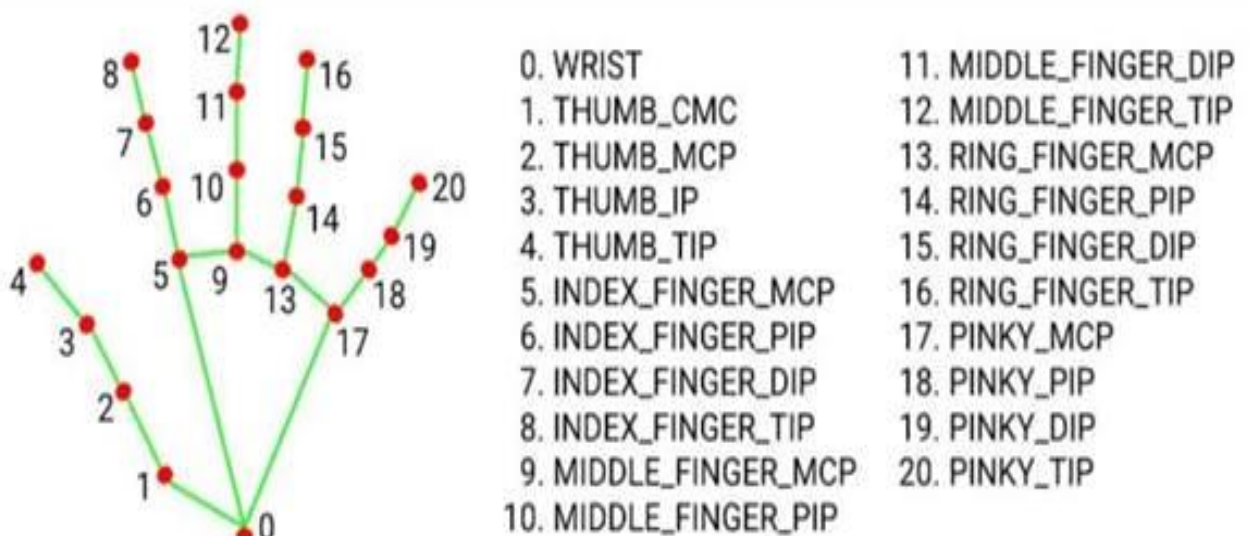
INTRODUCTION :

In this below program we use various in-built modules like *cv , mediapipe , pycaw , numpy* . In this program/project computer camera is used to process the input data to handle applications. The objective of this project is to develop an interface which will capture human hand gesture dynamically and will control the volume level . Libraries used:

- OpenCV is a great tool for image processing and performing computer vision tasks . It is an open-source library that can be used to perform tasks like *face detection , objection tracking , landmark detection, etc..*

- Mediapipe is a cross-platform library developed by Google that provides amazing ready-to-use Machine Learning solutions for [computer vision tasks](#).
- Pycaw is used to control the volume . The system requirements for it are [choco install visualcpp-build-tools](#).
- Numpy is used for data analysis tasks in Python. Numpy is mostly used for working with numerical values as it makes it [easy to apply mathematical functions](#).

Hand Landmark Model.



Algorithm:

- Detect Hand landmarks.
- Calculate the distance between thumb tip and index finger tip.
- Map the distance of thumb tip and index finger tip with volume range.
For my case, distance between thumb tip and index tip was within range of 70-200 and volume range was from -21.0 to +24.0 .

PROGRAM/CODE:

```
import cv2
import mediapipe as mp
import math
from ctypes import cast, POINTER
from comtypes import CLSCTX_ALL
from pycaw.pycaw import AudioUtilities, IAudioEndpointVolume
import numpy as np

devices = AudioUtilities.GetSpeakers()
interface = devices.Activate(IAudioEndpointVolume._iid_, CLSCTX_ALL, None)
volume = cast(interface, POINTER(IAudioEndpointVolume))
cap = cv2.VideoCapture(0)
mphands = mp.solutions.hands
hands = mphands.Hands()
mpdraw = mp.solutions.drawing_utils

while True:
    success, img = cap.read()
    imgrgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    results = hands.process(imgrgb)
    if results.multi_hand_landmarks:
        for handlms in results.multi_hand_landmarks:
            lmlist = []

            for id, lm in enumerate(handlms.landmark):
                h, w, c = img.shape
                cx, cy = int(lm.x * w), int(lm.y * h)
                lmlist.append([id, cx, cy])

            mpdraw.draw_landmarks(img, handlms, mphands.HAND_CONNECTIONS)

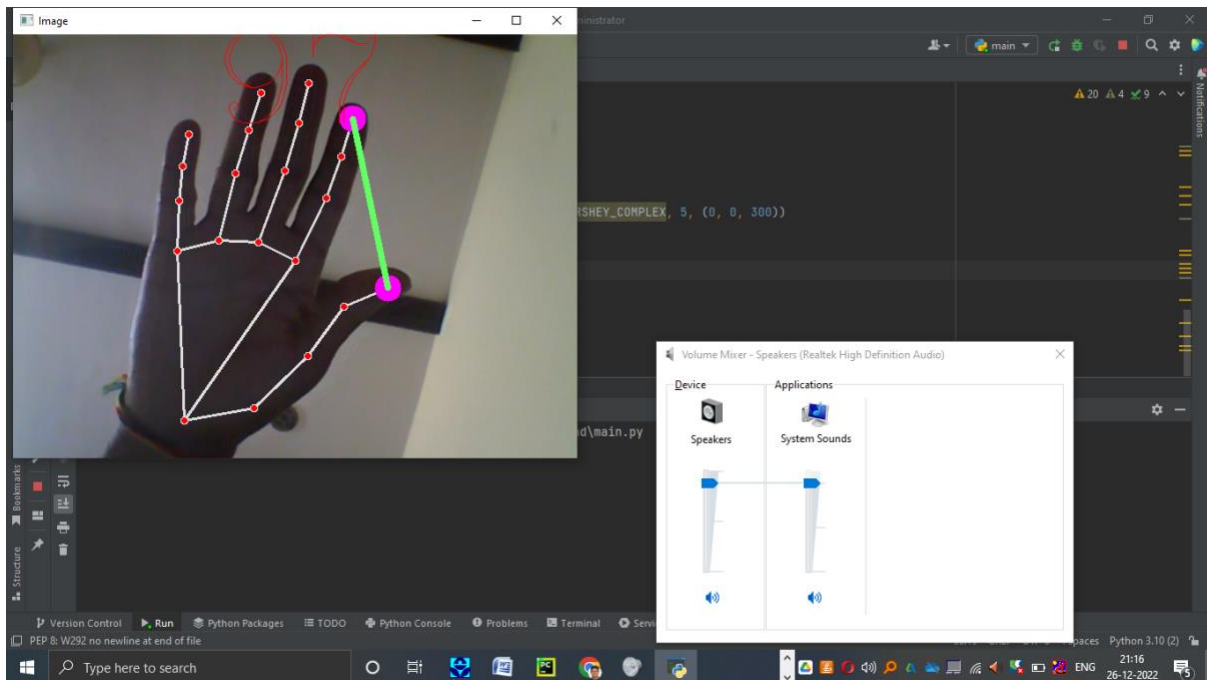
            if lmlist:
                x1, y1 = lmlist[4][1], lmlist[4][2]
                x2, y2 = lmlist[8][1], lmlist[8][2]
                cv2.circle(img, (x1, y1), 15, (255, 0, 255), cv2.FILLED)
                cv2.circle(img, (x2, y2), 15, (255, 0, 255), cv2.FILLED)
                cv2.line(img, (x1, y1), (x2, y2), (100, 255, 100), 5)
                len = math.hypot(x2 - x1, y2 - y1)

                if len < 70 or len > 200:
                    z1 = (x1 + x2) // 2
                    z2 = (y1 + y2) // 2
                    cv2.circle(img, (z1, z2), 15, (255, 0, 0), cv2.FILLED)

            volrange = volume.GetVolumeRange()
            minvol = volrange[0]
            maxvol = volrange[1]
            vol = np.interp(len, [70, 200], [minvol, maxvol])
            volume.SetMasterVolumeLevel(vol, None)
            volper = np.interp(len, [70, 200], [0, 100])
            cv2.putText(img, str(int(volper)), (225, 100),
                cv2.FONT_HERSHEY_COMPLEX, 5, (0, 0, 300))

    cv2.imshow("Image", img)
    cv2.waitKey(1)
```

OUTPUT:



CONCLUSION:

The program defines the sound control using hand and show the volume percentage of system in the image .Hence, the output defines the wireless sound control.